

Solutions to Mathematical Statistics (II) Final Exam.

Dec. 17, 2018

1.(350) Let $X_1, \dots, X_n$ be r.s. from a dist. with pdf $f(x : \theta) = \theta e^{-\theta x} I(x > 0)$.

(1)(40) Find the MLE $\hat{\theta}$ of θ , and find the approximate $100(1 - \alpha)\%$ C.I. for θ using the asymptotic normality of MLE.

(Sol) $l(\theta) = n \log \theta - \theta \sum X_i$, and $l'(\theta) = \frac{n}{\theta} - \sum X_i$. Hence, $\hat{\theta} = 1/\bar{X}$. Recall that $\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, I^{-1}(\theta))$. Now, $(\log f)'' = -1/\theta^2$, i.e., $\sqrt{n}(\frac{1}{\bar{X}} - \theta) \rightarrow N(0, \theta^2)$. Therefore, the approximate $100(1 - \alpha)\%$ C.I. for θ is $\frac{1}{\bar{X}} \pm z_{\alpha/2} \frac{1}{\sqrt{n\bar{X}}}$.

(2)(40) Compute $Var(\hat{\theta})$.

(Sol) Note that $X_1 \sim \Gamma(1, \theta^{-1})$, so that $Y \sim \Gamma(n, \theta^{-1})$. Now, $Var(\hat{\theta}) = n^2[E(Y^{-2}) - E^2(Y^{-1})]$, and it is easy to show that $E(Y^{-k}) = \theta^k \Gamma(n-k)/\Gamma(n)$. Hence, $Var(\hat{\theta}) = n^2 \theta^2 / (n-1)^2 (n-2)$.

(3)(40) Find the UMVUE $\tilde{\theta}$ of θ .

(Sol) Since $f(x : \theta) = \exp[-\theta x + \log \theta]$, $Y \equiv \sum X_i$ is CSS for θ . Now, $E(1/\bar{X}) = \theta/(n-1)$, so that $\tilde{\theta} = (n-1)/\sum X_i$.

(4)(40) Compute $Var(\tilde{\theta})$, and compare with $Var(\hat{\theta})$ and the Rao-Cramer lower bound.

(Sol) $Var(\hat{\theta}) = Var[(n-1)/Y] = \theta^2/(n-2)$ which is smaller than $Var(\hat{\theta})$. But, RC lower bound is $1/nI(\theta) = \theta^2/n$ which is smallest.

(5)(40) Consider a test of size α for testing $H_0 : \theta = 1$, vs $H_1 : \theta = 2$. Let C be a rejection region, $l(\theta, \delta)$ be a loss function with $l(1, 1) = l(2, 2) = 0$, $l(1, 2) = l(2, 1) = 1$, and let $R(\theta, \delta)$ be a risk function. Show that $R(1, C) = \alpha$ and $R(2, C) = \beta$, where β is a type II error.

(Sol) Let $L(\theta)$ be a likelihood. Then, $R(\theta, C) = \int_C l(\theta, 2)L(\theta) + \int_{C^c} l(\theta, 1)L(\theta)$ which implies that $R(1, C) = \int_C l(1, 2)L(1) + \int_{C^c} l(1, 1)L(1) = \int_C l(1, 2)L(1) = l(1, 2) \int_C L(1) = \alpha$. Similarly, $R(2, C) = \beta$.

(6)(50) Derive the LRT of size α for testing $H_0 : \theta = 1$, vs $H_1 : \theta \neq 1$.

(Sol) Let $Y = \sum X_i$, then $\frac{L(1)}{L(\hat{\theta})} = \frac{e^{-Y}}{(n/Y)^n e^{-(n/Y)Y}} = \frac{e^{-Y+n}}{n^n Y^{-n}} \leq c \Leftrightarrow y^n e^{-y} \leq d \Leftrightarrow Y \leq c_1$ or $Y \geq c_2$ because $y^n e^{-y}$ is maximized at $y = n$. Now, $Y \sim \Gamma(n, \theta^{-1}) \sim \Gamma(n, 1)$ under $H_0 : \theta = 1$. Hence, we have $2Y \sim \chi^2(2n)$. Therefore, LRT rejects H_0 if $2Y \geq \chi_{\alpha/2}^2$ or $2Y \leq \chi_{1-\alpha/2}^2$.

(7)(50) Show that the UMP test of size α for testing $H_0 : \theta = 1$, vs $H_1 : \theta \neq 1$ does not exist.

(Sol) Let $\theta' \neq 1$ and let $Y = \sum X_i$, then $\frac{L(1)}{L(\theta')} = \frac{e^{-Y}}{(\theta')^n e^{-\theta' Y}} = \theta'^{-n} e^{-(1-\theta')Y} \leq c \Leftrightarrow Y \leq c_1$ if $\theta' > 1$ and $Y \geq c_2$ if $\theta' < 1$. Therefore, the rejection region changes as either $\theta' > 1$ or $\theta' < 1$.

(8)(50) Let $s^2 = \sum (X_i - \bar{X})^2 / (n-1)$. Are $\bar{X}$ and $\bar{X}/s$ independent? Why or why not?

(Sol) Note that $f(x : \theta) = \theta e^{-\theta x} I(x > 0)$ is a scale family, and $\bar{X}$ is CSS for θ . Now, $\bar{X}/s$ is a scale-invariant statistic because $c\bar{X}/cs = \bar{X}/s$ for some $c \neq 0$ when we replace X_i by cX_i . Therefore, by the Basu's lemma, $\bar{X}$ and $\bar{X}/s$ are independent.

2.(150) Let $X_1, \dots, X_n$ be iid $N(0, \theta)$.

(1)(50) Find the UMVUE of $1/\theta$.

(Sol) Since the pdf is $\exp[-x^2/2\theta - \log(2\pi\theta)/2]$, it exp. fam. with $p(\theta) = -1/2\theta$, $K(x) = x^2$, so that $Y \equiv \sum X_i^2$ is CSS for θ . To find the UMVUE of $1/\theta$, note that $T \equiv Y/\theta \sim \chi^2(n)$. Hence, try $E(1/T) = \int_0^\infty t^{-1} t^{n/2-1} e^{-t/2} / \Gamma(n/2) 2^{n/2} dt = 1/(n-2)$. Therefore, $E((n-2)/Y) = 1/\theta$, so that $(n-2)/\sum X_i^2$ is the UMVUE of $1/\theta$.

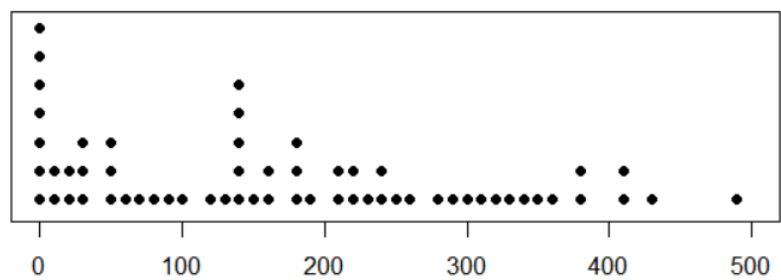
(2)(100) Compute the ARE of $\delta_{1n} = \sqrt{\pi/2} \sum |X_i|/n$ with respect to $\delta_{2n} = \sqrt{\sum X_i^2/n}$.

(Sol) Let $Y_i = \sqrt{\pi/2} |X_i|$, then $\delta_{1n} = \bar{Y}$. Then, by CLT, $\sqrt{n}(\bar{Y} - E(Y_1)) \rightarrow N(0, \text{Var}(Y_1))$. Now, $E(Y_1) = \sqrt{\pi/2} \int_{-\infty}^\infty |x| (2\pi\theta)^{-1/2} e^{-x^2/2\theta} dx$, which is equal to $\sqrt{\pi/2} \cdot 2 \int_0^\infty x (2\pi\theta)^{-1/2} e^{-x^2/2\theta} dx = \theta^{1/2}$, and $\text{Var}(Y_1) = E(\pi X^2/2) - \theta = (\frac{\pi}{2} - 1)\theta$. Hence, $\sqrt{n}(\delta_{1n} - \theta^{1/2}) \rightarrow N(0, (\frac{\pi}{2} - 1)\theta)$. For δ_{2n} , note that $Z_i \equiv X_i^2/\theta \sim \chi^2(1)$, so that $\sqrt{n}(\bar{Z} - 1) \rightarrow N(0, 2)$, i.e., $\sqrt{n}(\frac{1}{n} \sum \frac{X_i^2}{\theta} - 1) \rightarrow N(0, 2)$, i.e., $\sqrt{n}(\sum \frac{X_i^2}{n} - \theta) \rightarrow N(0, 2\theta^2)$. Now, by delta-method, $\sqrt{n}(\sqrt{\sum \frac{X_i^2}{n}} - \theta^{1/2}) \rightarrow N(0, \theta/2)$. Therefore, the ARE δ_{1n} with respect to δ_{2n} is $e_{1,2} = \frac{\theta/2}{(\frac{\pi}{2}-1)\theta} = 1/(\pi-2)$.

Summary Statistics for Final Exam. Scores

$n = 63$, $Ave. = 164$

Top Seat : 김승현 (490 / 500)



Summary Statistics for Total Scores

$n = 64$, $Ave. = 356$

Top Seat : 김승현 (930 / 1000)

